

M.A.R.I.A. - Status Report

Meta Analysis Recalibration Intelligence Architecture

Data: 2026-03-29 | Wersja: Cognitive Core K1-K13 + Stabilization + Faza F

1. Czym jest M.A.R.I.A.?

Lokalny, autonomiczny agent AI do samodzielnego uczenia sie z plikow tekstowych. Dziala na dedykowanym Mini PC (AMD Ryzen 5 7430U, 32GB RAM, Ubuntu 22.04). Offline-first, prywatny, bez chmury.

Początek projektu: 2025-11-14 **Deploy na produkcje:** 2026-02-22 **Testy:** 2448 passing **Sesji od narodzin:** ~1850+

2. Hardware i infrastruktura

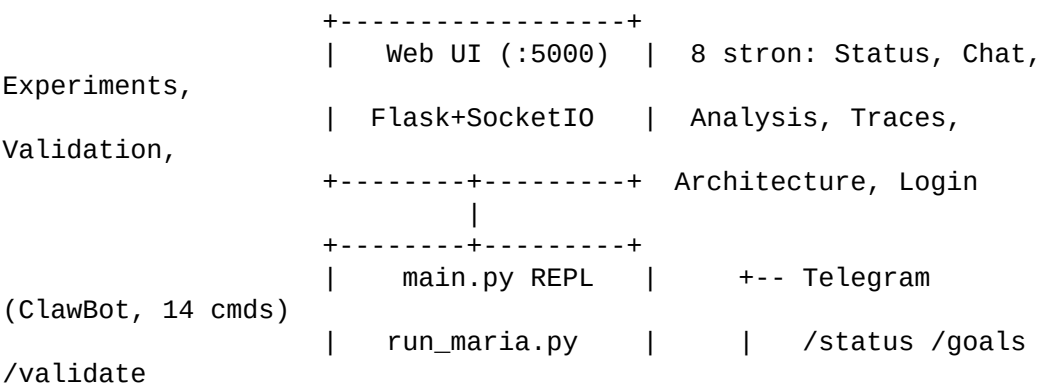
Komponent	Wartosc		----- -----	Mini PC	NiPoGi (AMD Ryzen 5 7430U, 32GB RAM, 1TB SSD)						
OS	Ubuntu 22.04 LTS			Storage	6TB ext4 dysk na logi/backupy/vision (/mnt/storage/)		Siec	LAN			
					192.168.178.32, WireGuard VPN		Bezpieczenstwo	UFW, fail2ban, SSH key-only, no root		Backup	
					Codziennie 3:00, 30 kopii na 6TB		Komunikacja	Telegram Bot (ClawBot) - 14 komend		Web UI	
					Flask+SocketIO :5000, PIN auth, 8 stron						

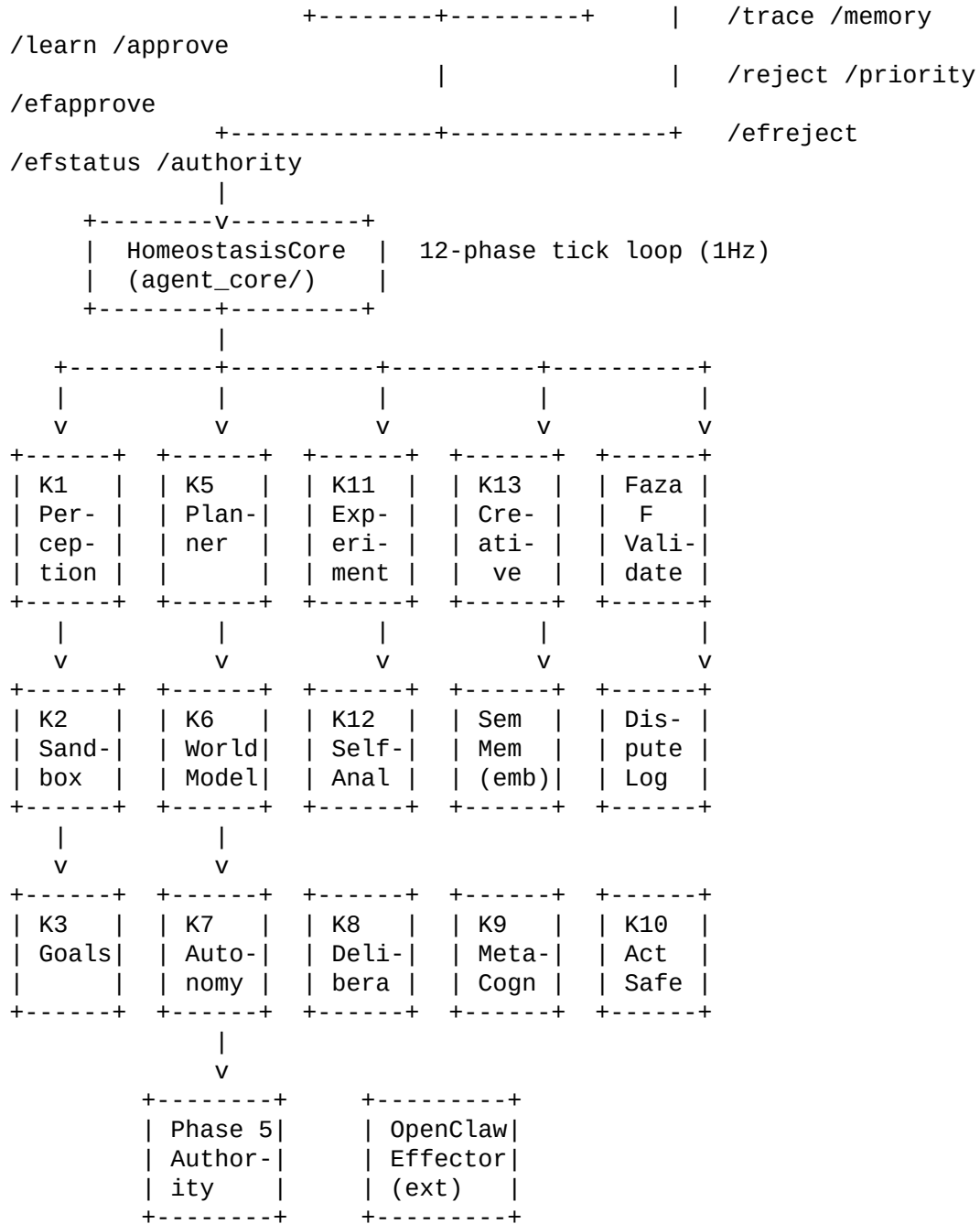
3. Multi-Model LLM Stack (7 modeli)

Model	Rola	RAM	Status		----- ----- ----- -----	llama3.1:8b	Executor / Brain (MODEL-02)	5GB	Warm	
(always loaded)										
	qwen3:8b	Strategic Planner (MODEL-01)	5.5GB	Cold (on-demand)			qwen2.5-coder:7b			
	Coder (MODEL-03)	5GB	Cold (on-demand)			Rule-based	Triage (MODEL-04)	0GB	Keyword classifier	
	nomic-embed-text	Embeddings (MODEL-05)	274MB	Cold (on-demand)			z-ai/glm5 (NIM)	Cloud API		
	(MODEL-06)	0GB	40 RPM, expiry Aug 2026			Codex CLI (ChatGPT)	Encyclopedia (MODEL-07)	0GB	10	
	calls/h					qwen2.5:3b	OpenClaw Effector	2GB	Separate instance	

Heavy mutex: MODEL-01 i MODEL-03 nigdy jednocześnie (RAM guard). **Golden rule:** MODEL-02 warm, reszta on-demand.

4. Architektura modułow





5. Cognitive Core - Kontrakty K1-K13 (COMPLETE)

| K# | Nazwa | Opis | Testy | |----|-----|-----|-----| | K1 | Unified Perception | PerceptionEvent, Buffer, 6 adapterow | 131 | | K2 | Sandbox/Production | SandboxManager, transaction log, recovery | 44 | | K3 | Goal System | 4 typy celow, 6 statusow, PROPOSED flow | 63 | | K4 | Evaluation | READ-ONLY observer, 5 metryk | 35 | | K5 | Planner | ReAct loop, 13 action types, hybrid frequency | 136 | | K6 | World Model | BeliefStore (JSONL, MERGE, cap 2000) | 69 | | K7 | Autonomy Policy | FREE/GUARDED/RESTRICTED/FORBIDDEN + Phase 5 authority | 45 | | K8 | Deliberation | 3 strategy templates, IntentTracker | 49 | | K9 | Meta-Cognition | ReflectionStore, ConfidenceTracker, needs_human() | 73 | | K10 | Action Safety | SafetyMode(3), AuditLog, EffectValidator | 52 | | K11 | Experiments |

ProposalEngine, ParameterRegistry, ADOPT/REJECT | 67 | | K12 | Self-Analysis | NIM cascade analyzer,
PROPOSED goals | 45 | | K13 | Creative Module | TensionDetector, NIM engines, meta-goals | 130 |

6. Stabilization Roadmap (6 faz - COMPLETE)

| Phase | Nazwa | Opis | ADR | |-----|-----|-----|-----| | 1 | Decision Traceability | Episode-based traces, correlation
IDs | ADR-022 | | 2 | Memory Consistency | MemoryQuery API, staleness fixes, grounding | ADR-023 | | 3 |
Scheduler Hardening | Execution budgets, Ollama timeouts, degradation | ADR-024 | | 4 | Autonomy Governance |
Cross-metric validation, promotion audit | ADR-025 | | 5 | Effector Safety | 5-level authority, approval queue,
anti-cascade | ADR-026 | | 6 | Readiness Review | 100-cycle marathon, 15-point checklist | - |

All 5 gates passed: Gate A (tracing) -> Gate B (memory) -> Gate C (budgets) -> Gate D (governance) -> Gate E (readiness).

7. Faza F: Multi-Source Learning (COMPLETE)

Maria uczy sie z jednego LLM (Ollama), potem drugi LLM (NIM) niezaleznie analizuje ten sam material. Roznice logowane jako spory.

| Komponent | Opis | |-----|-----| | CrossValidator | Porownanie primary (Ollama) vs secondary (NIM) | |
ConfidenceScorer | Rule-based scoring (Jaccard similarity, 3 wymiary) | | DisputeLog | JSONL persistence sporow,
thread-safe | | Planner trigger | _maybe_validate() co 6h, K7 GUARDED | | Belief update | OBSERVATION->FACT
(>0.7), demotion HYPOTHESIS (<0.3) | | Web UI | /validation page (stats, disputes, history) | | Telegram | /validate [disputes|unresolved] |

8. Homeostasis - 12-fazowy tick loop

Phase 1: SENSE (5 sensors: resource, cognitive, thermal, power, time)
Phase 2: INTERPRET (raw -> semantic state)
Phase 3: VALIDATE (constraints + CRITICAL alarms)
Phase 4: DECIDE MODE (ACTIVE -> REDUCED -> SLEEP -> SURVIVAL)
Phase 5: GENERATE CORRECTIVE ACTIONS
Phase 6: EXECUTE CORRECTIVE ACTIONS
Phase 7: UPDATE HEALTH SCORE
Phase 8: PERCEIVE (K1 - aggregate events -> PerceptionBuffer)
Phase 9: TEACHER TRIGGER (idle >= 10min -> auto-learn)
Phase 9.5: MODEL SCHEDULER (load/unload, RAM guard, heavy mutex)
Phase 10: PLANNER (K5-K13: guard -> select -> plan -> execute -> trace)
Phase 11: TELEGRAM (poll co 30s, operator commands)
Phase 12: AUDIT & LOG

9. Telegram (ClawBot) - 14 komend

| Komenda | Opis | |-----|-----| | /status | Stan systemu (mode, health, planner, knowledge) | | /goals | Lista celow
(active + proposed, ID, priority) | | /validate | Cross-validation stats | | /validate disputes | Ostatnie spory | | /trace [N|
stats|failed|ep-ID] | Decision traces | | /memory <temat> | Co Maria wie o temacie | | /memory gaps | Luki w wiedzy |
| /learn <temat> | Naucz sie o temacie | | /approve <id> | Zatwierdz proposed goal | | /reject <id> | Odrzuc proposed

goal || /priority <id> <0-1> | Zmien priorytet celu || /efapprove <id> | Zatwierdz akcje efektora || /efreject <id> | Odrzuc akcje efektora || /efstatus | Status efektora i approval queue || /authority [level] | Zmien poziom autoryzacji (observe/confirm/bounded) || /restart | Restart Marii (systemd wskrzesi po 10s) |

10. Web UI (8 stron)

| Strona | URL | Opis | |-----|-----|-----| | Status | /status | 8-panel Metaoperator dashboard || Chat | / | WebSocket chat z Maria || Experiments | /experiments | K11 propozycje, raporty, parametry || Analysis | /analysis | K12 self-analysis raporty || Traces | /traces | Decision traces (Phase 1) || Validation | /validation | Cross-validation stats + disputes (Faza F) || Architecture | /architecture | Force graph + pipeline + data flow || Login | /login | PIN authentication |

11. Stabilnosc produkcyjna

| Metryka | Wartosc | |-----|-----| | RAM (Maria process) | ~67MB RSS || Health score | 0.84-0.99 || OOM crashes since deploy | 0 (fix 2026-03-18) || Uptime (typical) | 24/7 z systemd auto-restart || Ollama timeout | 120-180s per role (Phase 3) || Episode budget | max 10 LLM calls, 5min latency || Authority default | OBSERVE (safe, backward compatible) |

12. Decyzje architektoniczne (ADR)

26 ADR zarejestrowanych (ADR-001 do ADR-027), pelna lista w docs/ROADMAP.md.

Kluczowe:

- **ADR-001:** JSONL jako source of truth
 - **ADR-013:** Planner rule-based (zero LLM, deterministic)
 - **ADR-014:** Najpierw mozg, potem zmysly
 - **ADR-015:** Multi-organ model stack (heavy mutex)
 - **ADR-021:** Semantic Memory via embeddings
 - **ADR-026:** Effector safety envelope (5-level authority)
 - **ADR-027:** Multi-Source Learning via post-learning cross-validation
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13. Co dalej

| Priorytet | Opis | Status | |-----|-----|-----| | Belief Store compaction | Compaction oparta na waznosc, dedup | PLANNED || CDL dopracowanie | Lepsze rozpoznawanie intencji, feedback loop | PLANNED || Web UI polish | Dense mode, sidebar | PLANNED || Vision (Warstwa 10) | Kamera Tapo C200 z RTSP | CZEKA NA SPRZET || Smart Home (Warstwa 11) | Shelly/Tasmota IoT | CZEKA NA SPRZET |

Wygenerowano: 2026-03-29 Testy: 2448 passing | Kontrakty: K1-K13 COMPLETE | Stabilization: 6/6 COMPLETE | Faza F: COMPLETE